

TECH-ARCH-4184: Architecture Reference Document

Document Metadata

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1. Document Scope and Executive Summary

This Architecture Reference Document (ARD) outlines the core policies, standard operating procedures, and guidelines for the Architecture department.

The scope of this ARD includes:

- * Defining architecture principles and guidelines
- * Establishing standards for design, development, and testing
- * Describing the roles and responsibilities of architecture team members
- * Outlining compliance monitoring and penalty procedures

2. Core Policies and Standard Operating Procedures

2.1 Architecture Principles

The following principles guide the Architecture department's decision-making processes:

- * Separation of Concerns (SoC): each component has a single, well-defined responsibility
- * Single Responsibility Principle (SRP): each component is responsible for only one thing
- * Don't Repeat Yourself (DRY): avoid duplication of code and effort
- * KISS: keep it simple and straightforward

2.2 Design Guidelines

The Architecture department follows these design guidelines:

- * Use a modular architecture with clear boundaries between components
- * Implement a scalable and fault-tolerant system
- * Follow a consistent naming convention for variables, functions, and classes
- * Use a version control system to track changes and maintain a clean history

2.3 Development Process

The development process involves the following steps:

1. Requirements gathering and analysis
2. Design creation and review
3. Implementation and testing
4. Deployment and maintenance

2.4 Testing and Quality Assurance

The Architecture department conducts the following types of testing:

- * Unit testing: individual components are tested in isolation
- * Integration testing: multiple components are tested together
- * System testing: the entire system is tested as a whole
- * User acceptance testing (UAT): users test the system to ensure it meets their needs

3. Compliance Monitoring, Penalties, and Operational Governance

3.1 Compliance Monitoring

The Architecture department will:

- * Conduct regular code reviews to ensure adherence to design guidelines and best practices
- * Monitor system performance and scalability to identify areas for improvement
- * Track changes and updates made to the architecture and infrastructure

3.2 Penalties and Consequences

In case of non-compliance, the following penalties may apply:

- * Verbal warning: first-time offenders receive a verbal warning
- * Written warning: repeat offenders receive a written warning
- * Suspension or termination: severe or repeated offenses result in suspension or termination of architecture team member privileges

3.3 Operational Governance

The Architecture department will maintain accurate and up-to-date records of:

- * Design decisions and rationale
- * System performance metrics and scalability data
- * Compliance monitoring results and penalties issued

By following this Architecture Reference Document, the Architecture department ensures that software systems are designed and